

Algorithm

1. Start the program.
2. Read a string from the user.
3. Remove the newline character inserted by `fgets()`.
4. Display the menu:
 - Count characters.
 - Print each character on a new line.
 - Count vowels and consonants.
 - Exit.
5. Read the user's choice.
6. Use `switch` to perform the selected operation.
7. For **Choice 1**, traverse the string until `'\0'` and count each character.
8. For **Choice 2**, traverse the string and print every character on a separate line.
9. For **Choice 3**:
 - Traverse each character.
 - Convert uppercase letters to lowercase for comparison.
 - If the character is `a, e, i, o, u`, increment the vowel count.
 - If it is another alphabetic character, increment the consonant count.
 - Ignore spaces, digits, and special characters.
10. For **Choice 4**, terminate the program.
11. For any other choice, display an error message.
12. Repeat the menu until the user selects Exit.
13. Stop.

Test Case Table

Test	Input String	Choice	Expected Result
1	Hello	1	Number of characters = 5
2	CAT	2	C, A, T printed on separate lines
3	Hello World	3	Vowels = 3, Consonants = 7
4	AEIOU	3	Vowels = 5, Consonants = 0
5	bcd fg	3	Vowels = 0, Consonants = 5
6	C Programming 123	3	Vowels = 3, Consonants = 9
7	Hello	4	Program exits
8	Hello		

C Program

```
#include <stdio.h>
```

```
int countCharacters(char str[])  
{
```

```

    int count = 0;

    while (str[count] != '\0')
    {
        count++;
    }

    return count;
}

void printCharacters(char str[])
{
    int i = 0;

    printf("\nCharacters in the string:\n");

    while (str[i] != '\0')
    {
        printf("%c\n", str[i]);
        i++;
    }
}

void countVowelsConsonants(char str[])
{
    int i = 0;
    int vowels = 0, consonants = 0;
    char ch;

    while (str[i] != '\0')
    {
        ch = str[i];

        if (ch >= 'A' && ch <= 'Z')
        {
            ch = ch + 32;
        }

        if (ch >= 'a' && ch <= 'z')
        {
            if (ch == 'a' || ch == 'e' || ch == 'i' ||
                ch == 'o' || ch == 'u')
            {
                vowels++;
            }
        }
    }
}

```

```

        else
        {
            consonants++;
        }
    }

    i++;
}

printf("Number of vowels = %d\n", vowels);
printf("Number of consonants = %d\n", consonants);
}

int main()
{
    char str[100];
    int choice;
    int length;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);

    /* Remove newline added by fgets */
    int i = 0;
    while (str[i] != '\0')
    {
        if (str[i] == '\n')
        {
            str[i] = '\0';
            break;
        }
        i++;
    }

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Count characters in the string\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)

```

```

        {
            case 1:
                length = countCharacters(str);
                printf("Number of characters = %d\n",
length);
                break;

            case 2:
                printCharacters(str);
                break;

            case 3:
                countVowelsConsonants(str);
                break;

            case 4:
                printf("Exiting the program...\n");
                break;

            default:
                printf("Invalid choice. Please enter a choice
from 1 to 4.\n");
        }

    } while (choice != 4);

    return 0;
}

```

Execution

```
Last login: Tue Sep 29 22:33:41 on ttys000
[bhavya@Bhavyas-MacBook-Air ~ % nano day36.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day36.c -o day36
[bhavya@Bhavyas-MacBook-Air ~ % ./day36
Enter a string: Hello world

----- MENU -----
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 3
Number of consonants = 7

----- MENU -----
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting the program...
bhavya@Bhavyas-MacBook-Air ~ % █
```